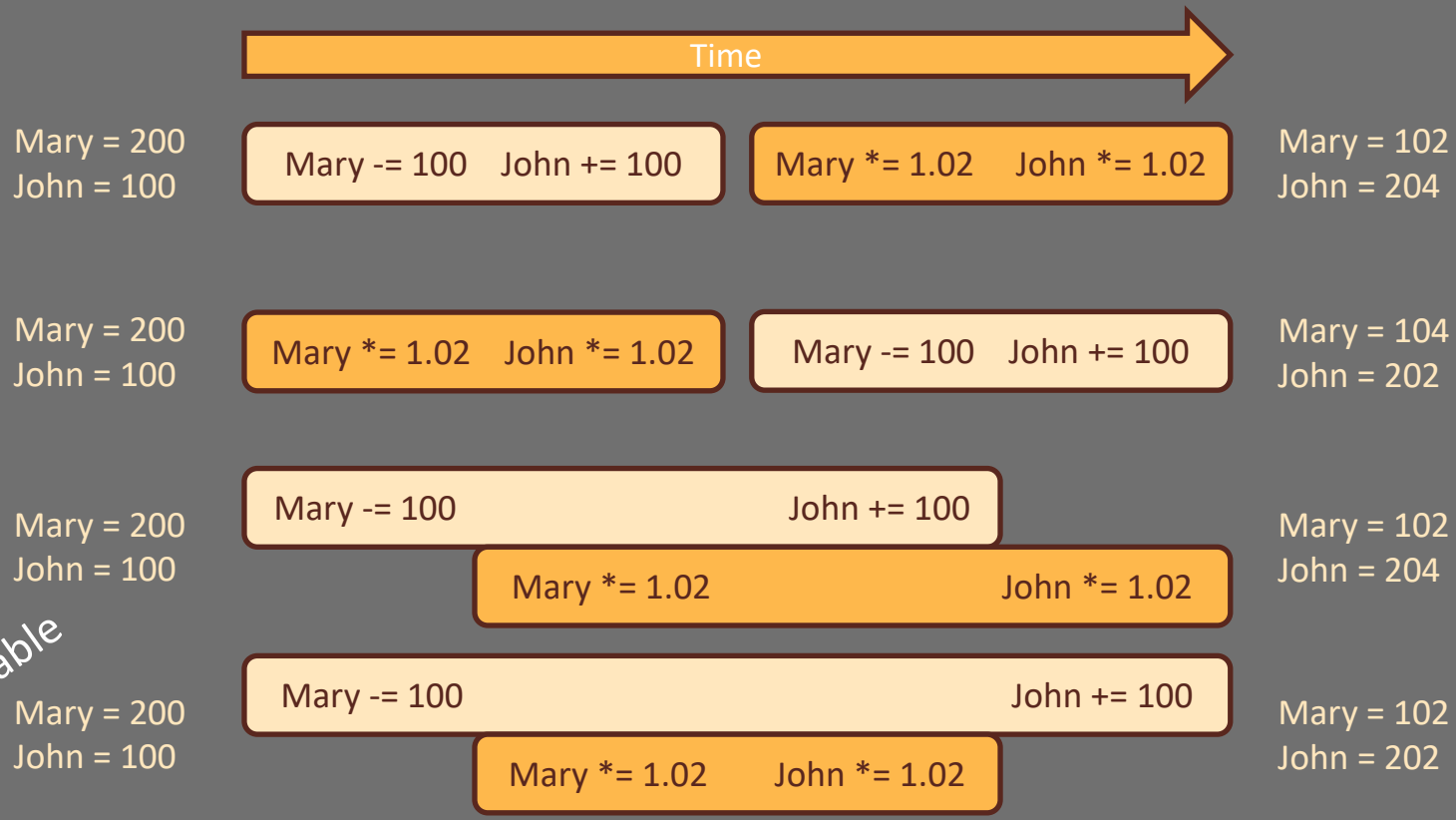


Schedules

Schedules


Time
Mary = 200
John = 100
Mary -= 100
John += 100
Mary *= 1.02
John *= 1.02
Mary = 102
John = 204
Mary = 200
John = 100
Mary *= 1.02
John *= 1.02
Mary -= 100
John += 100
Mary = 104
John = 202
Serializable
Mary = 200
John = 100
Mary -= 100
John += 100
Mary *= 1.02
John *= 1.02
Mary = 102
John = 204
Non-serializable
Mary = 200
John = 100
Mary -= 100
John += 100
Mary *= 1.02
John *= 1.02
Mary = 102
John = 202

Schedule terminology

	Meaning
Serial	No interleaving
X,Y **equivalent**	**For any DB state**, effect of X is identical to effect of Y.
Serializable	Equivalent to **some** serial schedule.

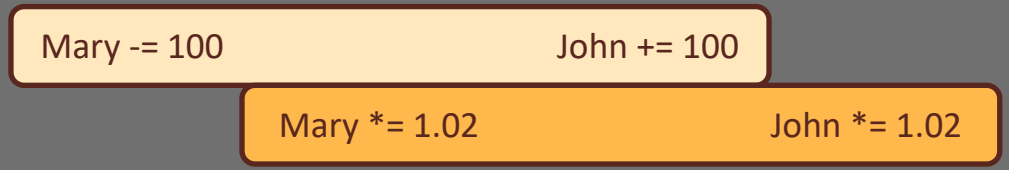
Testing serializability is expensive

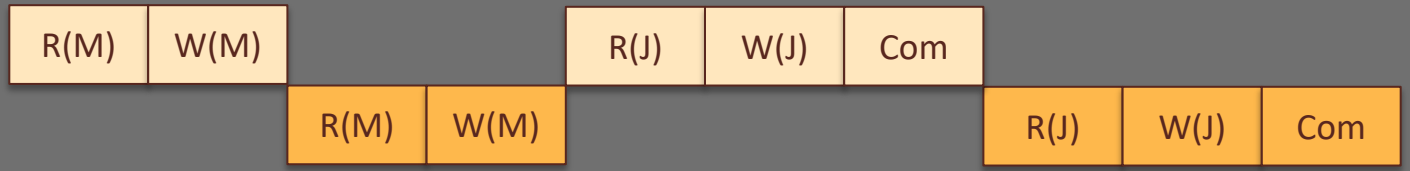
n transactions of lengths $l_1, \ldots, l_n$

- # serial schedules? $n!$
- # schedules? $\binom{\sum l_i}{l_1, \ldots, l_n} = \frac{(\sum l_i)!}{\prod l_i!}$

Simplified schedules

- Pay attention to reads and writes.
- Ignore the specific computations.





Road map

- Conflicts in schedules
- Conflict serializability $\rightarrow$ serializability

- Anomalies in schedules
- Isolation levels control which anomalies are allowed

- Locking to implement schedule